Automatic Pet Feeder

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***Abstract*—This report details the design and implementation of an automatic pet feeder with remote control functionality. The feeder employs a load sensor to determine food presence in the bowl and a stepper motor with a paddle mechanism for controlled food dispensing. A Node.js server facilitates communication between the web interface and the Raspberry Pi using the MQTT protocol. This system offers precise feeding control, user convenience, and real-time monitoring capabilities, promoting a healthy and consistent diet for pets.**

***Keywords*—Automatic pet feeder, remote control, sensor, stepper motor, Node.js, MQTT, web interface**

I. Introduction

Diet is not only important for humans but also critical for pets at home. Research shows that many diseases in pets are caused by poor dietary habits [1]. An automatic pet feeder with remote control functionality can precisely control pets' diets and ensure they are fed on time, even when no one is nearby [1].

The group creates a smart pet feeder equipped with a loading sensor and a stepping motor with a paddle connected to the end of the motor to regulate the amount of food dispensed into the feeding bowl. The feeding order will be given remotely by the user, and once the smart pet feeder has received the feeding order, the load sensor will determine if there is sufficient food already remains in the feeding bowl. If so, the stepping motor will not be activated, otherwise, the smart pet feeder will deploy the feeding mechanism. The feeding mechanism involves placing the food reservoir directly above the feeding bowl, with the paddle at the bottom of the reservoir to control the flow of food into the bowl. When the stepping motor is activated, it rotates the paddle and allows the food to pour into the feeding bowl by gravity.

We also implement a website for users to send commands and check the past feeding activities. The server implementation involves creating a Node.js-based web server which interacts with MQTT broker to manage and log commands sent to Raspberry Pi.

II. Innovative Design for Automated Pet Feeder Mechanism

At the beginning of the project, the team considered using an electronically controlled valve to regulate the amount of food dispensed into the feeding bowl. However, implementing the electronic valve alongside the pet food container posed significant challenges. The shapes of both the valve and the food container would need to be specifically designed to work in harmony, complicating the system [2]. To simplify the feeding mechanism, a cross-shaped paddle, as depicted in *Fig. 1* and *Fig. 2*, was 3D printed and connected to a stepping motor.

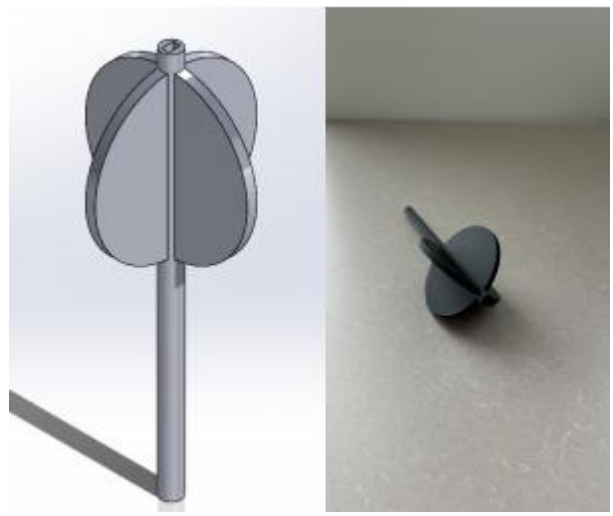


Fig. 1 and *Fig. 2* 3D printed cross-plane shaped paddle

The chosen design features a cylindrical container as the pet food reservoir. The paddle's dimensions are specifically tailored to match the container's radius [1]. In its default state, the paddle's blades prevent food from falling into the feeding bowl. When the stepping motor is activated, it rotates the paddle, allowing food from the container to be released into the bowl. This mechanism is positioned directly above the feeding bowl at an appropriate height, ensuring that when the feeding process is triggered, all the food is accurately dispensed into the bowl [3].

To enhance the system's efficiency, a load sensor is placed at the bottom of the feeding bowl. This sensor measures the weight exerted on the bowl to determine whether there is sufficient food present. If the sensor detects a certain threshold of weight, it indicates that the bowl already contains enough food. In such cases, even if a feeding command is issued, the feeding process will not be activated, preventing overfeeding and ensuring a consistent and controlled food supply for the pet.

The innovative approach of using a paddle mechanism instead of an electronically controlled valve simplifies the design and operation of the feeder. It ensures precise control

over food dispensing while minimizing the complexity of integrating various components, making the automated pet feeder both effective and reliable.

III. Components and Connections

In the automated feeding system, two critical components — the load sensor and the stepper motor — play essential roles in ensuring precise and reliable operation. The system begins with the integration of a battery power board onto the breadboard, providing a stable and independent power source for all components. This setup is crucial for maintaining consistent performance and avoiding power fluctuations that could affect the system's accuracy.

The load sensor chosen for this application is the RP-S40-ST, a high-accuracy thin film pressure sensor with a measurement range of 20g to 10kg (as shown in Fig. 3) [4]. This sensor is ideal for the feeding system due to its ability to accurately detect small changes in weight, which is essential for monitoring the quantity of food in the feeding bowl. The RP-S40-ST's compact and thin design allows it to fit seamlessly into the system, making it a practical choice for space-constrained environments.

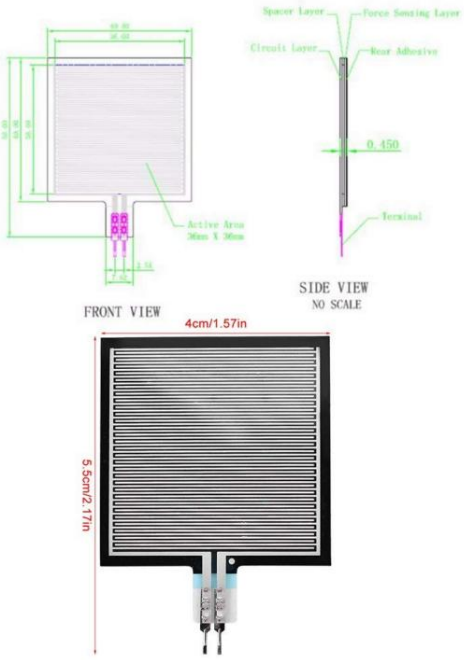


Fig. 3 RP-S40-ST 20g-10kg High Accuracy Thin Film Pressure Sensor, Force Sensor for Intelligent High-End Seat

The sensor can be directly attached to the breadboard, simplifying the installation and integration process [4]. It is powered by the battery board, ensuring that it operates independently of the Raspberry Pi's power supply. This independence is critical for providing consistent performance even during power fluctuations that may affect the Raspberry Pi. The sensor sends force data to the Raspberry Pi via a connection to one of its GPIO pins. This data is crucial for real-time monitoring of the food's weight, enabling the system to adjust the feeding process as necessary. By continuously tracking the food quantity, the system ensures that the pet receives the correct amount of food, preventing issues like overfeeding or underfeeding.

The stepper motor setup in the automated feeding system requires a ULN2003 driver, which acts as an interface between the motor and the Raspberry Pi. The motor, equipped with a connector that fits directly into the ULN2003 driver board, is straightforward to integrate into the system. Once the motor is secured to the ULN2003, the next step is to connect the driver board to the Raspberry Pi using jumper wires. The input pins of the ULN2003 driver board need to be connected to the corresponding GPIO pins on the Raspberry Pi, facilitating the control of the motor.

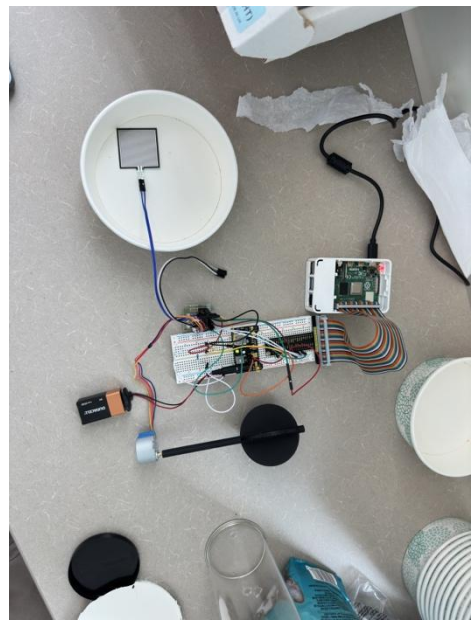


Fig. 4 Load Sensor and Stepper Motor in Automated Feeding System

It is crucial to ensure the stepper motor receives sufficient power to maintain its performance [5]. To achieve this, the VCC and GND of the ULN2003 board are connected to the battery power board. This connection ensures that the stepper motor operates reliably and consistently, independent of the limited current provided by the Raspberry Pi.

By carefully integrating these components, the automated feeding system can effectively manage the feeding process. This ensures precise control over food dispensing and continuous monitoring of the food quantity in the bowl through the load sensor. This integrated approach provides an efficient feeding solution and enhances the system's reliability and functionality. It ensures the well-being of the pet by maintaining a consistent and appropriate feeding regimen, contributing to a balanced and healthy diet.

IV. Feeding Algorithm

The feeding algorithm is designed to ensure a reliable and efficient feeding process. The process is initiated when a user sends a command through the web interface. This command is then transmitted to the Raspberry Pi via an MQTT broker [2]. Upon receiving the command, the Raspberry Pi verifies its authenticity and logs it with a timestamp.

The system then utilizes a load sensor located at the bottom of the feeding bowl to measure the current weight of the food. The sensor data is transmitted to the Raspberry Pi for analysis. If the weight exceeds a predetermined threshold, indicating that there is already enough food in the bowl, the feeding process is terminated to prevent overfeeding[3].

However, if the food weight is below the threshold, the Raspberry Pi activates a stepper motor. This motor rotates a paddle mechanism, facilitating a controlled amount of food to flow from the reservoir into the feeding bowl.

Following food dispensing, the load sensor re-evaluates the weight in the bowl. The updated weight is logged and compared to ensure the correct amount of food has been dispensed.

In the event of any discrepancies or errors during the feeding process, such as a motor malfunction or sensor failure, the system logs the error and notifies the user through the web interface [2].

Upon successful completion of the feeding process, the system logs the event with a timestamp, and the feeder reverts to standby mode, ready for the next command.

V. Comprehensive Server for Feeding Orders

The feeding algorithm ensures a reliable and efficient feeding process for the automatic pet feeder. It involves multiple steps, from user command initiation to post-feeding verification. Below is the detailed feeding logic:

To comprehensively manage the communications and user data for the smart pet feeder. A server is built by using Express which is a web framework in Node.js and integrates with the MQTT protocol to facilitate communication between the web interface and the Raspberry Pi. Below is the detailed summary of server set-up:

A. Setting up the server

We began by setting up a Node.js environment and installing necessary packages such as express for the web server, mqtt for MQTT communication, fs for file system operations, and moment-timezone for time zone management [2].

The server will be host on port 3000 and static file are under public directory

B. MQTT Integration

The server connects to an MQTT broker (in this case, broker.hivemq.com) using the mqtt library [2].

It subscribes to the raspberry/logs topic to receive log messages from the Raspberry Pi.

The on('message') event handler processes incoming MQTT messages, appending them to a log.csv file for record-keeping.

C. Commands

The server defines an endpoint /send_command/:command that allows the web interface to send commands to the Raspberry Pi via MQTT.

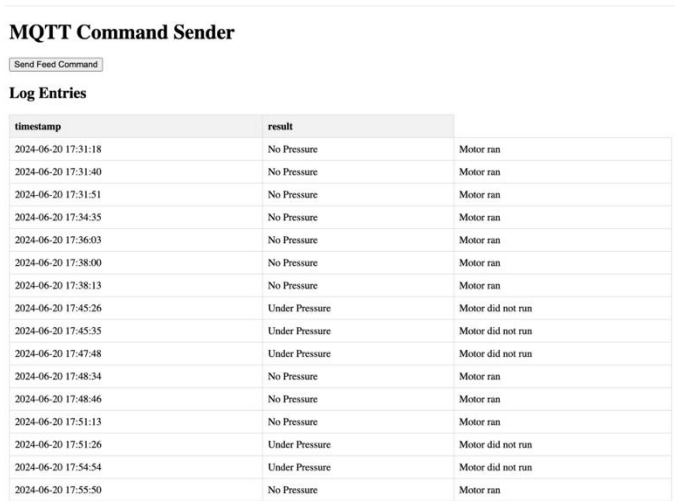
When a command is received, it is timestamped in the Canada/Hamilton time Zone using the moment-timezone library and then published to the raspberry/command topic [2].

The command and its timestamp are also appended to the log.csv file for logging purposes.

D. Web Interface

The web interface, served from the public directory, includes an HTML file (index.html) that provides a user-friendly interface to send commands and view log entries.

JavaScript within the HTML file fetches the log data from the server and displays it in a table [2].



MQTT Command Sender

Send Feed Command

Log Entries

timestamp	result	
2024-06-20 17:31:18	No Pressure	Motor ran
2024-06-20 17:31:40	No Pressure	Motor ran
2024-06-20 17:31:51	No Pressure	Motor ran
2024-06-20 17:34:35	No Pressure	Motor ran
2024-06-20 17:36:03	No Pressure	Motor ran
2024-06-20 17:38:00	No Pressure	Motor ran
2024-06-20 17:38:13	No Pressure	Motor ran
2024-06-20 17:45:26	Under Pressure	Motor did not run
2024-06-20 17:45:35	Under Pressure	Motor did not run
2024-06-20 17:47:48	Under Pressure	Motor did not run
2024-06-20 17:48:34	No Pressure	Motor ran
2024-06-20 17:48:46	No Pressure	Motor ran
2024-06-20 17:51:13	No Pressure	Motor ran
2024-06-20 17:51:26	Under Pressure	Motor did not run
2024-06-20 17:54:54	Under Pressure	Motor did not run
2024-06-20 17:55:50	No Pressure	Motor ran

Fig. 5 Graphic user interface

This comprehensive setup ensures seamless communication between the web interface and the Raspberry Pi, reliable logging of commands and their outcomes, and proper time zone management for accurate record-keeping. The server facilitates command transmission and provides a robust logging mechanism easily accessible via a web interface (*Fig. 5*).

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